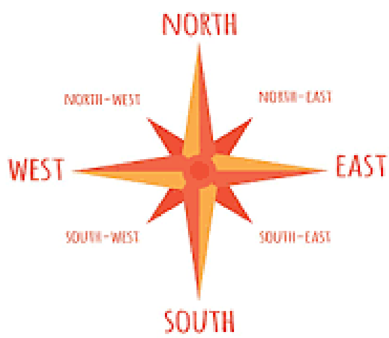
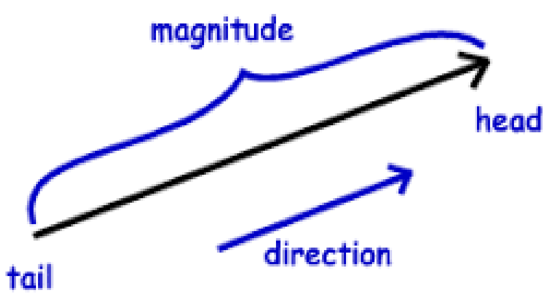
Unit 1: Vectors





In Physics, there are two types of quantities: vectors and scalars. Many of the topics covered in this course will involve vectors.

This includes topics such as forces, velocities, accelerations, and more. Being able to visualize a vector and properly break it down into its components is critical to understanding any physics problem we might encounter this year. It is usually a safe bet to assume that when you come across a vector quantity, there is a 50/50 change you will need to break it down into its components.

What is a vector?

As previously mentioned, there are two types of quantities: scalar and vector. To better understand them, we first need to know what magnitude and direction are. Magnitude is the size of something. Think about earthquakes. A 0.5 magnitude earthquake is a small earthquake that doesn't impact a large area. A 7.5 magnitude earthquake is a very large earthquake that impacts a very large area. So the magnitude of something looks at its size. In physics, magnitude can tell us if something is a strong or weak force, fast or slow speed, and much more. The direction of something is exactly what is sounds like. Is something traveling up, down, left, or right. To make things more uniform, physics uses compass directions of North, South, East, and West.

A **scalar** quantity is a quantity that only includes magnitude but does not look at direction. On the other hand, a **vector** quantity is a quantity that includes both magnitude and direction. Some examples of scalar quantities include:

- Temperature: 45° F includes a magnitude of 45 but does not include a direction
- Distance: 89 meters
- Time: 35 seconds has a magnitude of 35 but does not include a direction
- Speed: 137 mph has a magnitude of 137 but does not include a direction
- Volume: A 10 gallon container has a magnitude of 10 but no direction

Notice how all scalar quantities have a set number attributed to them as magnitudes but there is no direction attached. It makes sense for most of these quantities. You don't see the Weather apps telling you that it is 95 degrees South outside or your timer being set for 15 minutes West. ***"What about 137 mph east?" We will talk about why speed is scalar and not vector in the upcoming paragraphs*** Now, lets turn our attention to vector quantities. Some examples of vector quantities include:

- Force: When a force is acting on something, there is a direction. Let's say I push on as desk with a 5N West force This has both a magnitude (5N) and direction (West). Every force that has direction. This can be up, down, left, right, or somewhere in between.
- Acceleration: When something accelerates it has both magnitude and direction. If you are on King Da Ka, you are accelerated at $10m/s^2$ due North (aka straight ahead). This acceleration has both a magnitude of 10 and a direction of North. Imagine if you were given an acceleration of $10m/s^2$ South and launched back into the station or $10m/s^2$ East into the trees. As you can see, direction is very important when looking at vector quantities.
- Velocity: As promised, we will now learn why speed is always scalar. Speed is scalar because it has no direction attached. The minute you attach direction to the rate something is traveling, it becomes velocity. 15m/s is scalar and somethings speed. 15m/s due East is a vector. Velocity is essentially speed (a scalar) with direction attached (making it a vector).

Obviously, we will not be going into formal definitions of forces, accelerations, velocities, and all of the other possible scalars and vectors we will encounter in physics. All you need to get from this is the difference between scalars and vectors.

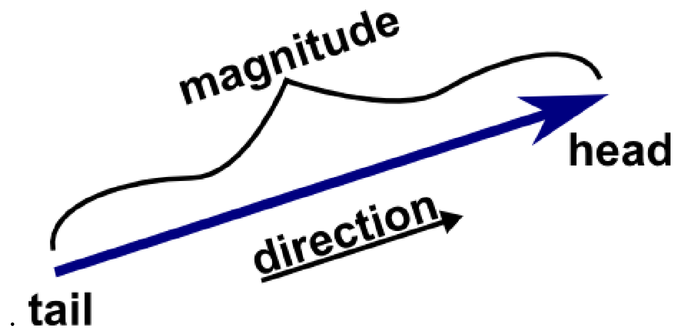
Vectors vs Scalar Practice Problems:

Write if each of the following examples is a vector quantity or a scalar quantiy.

1. A runner running for 3 hours ___________________________
2. A gas tank containing 20 gallons ___________________________
3. The work a crane does to lift a container 50 feet off the ground ______________________
4. A pitcher throwing a ball to the catcher with a 65 N force ___________________________
5. A car traveling with a speed of 45 mph ___________________________
6. A storm with wins blowing 120 mph Northeast. ___________________________
7. The freezing point of water at 32° F ___________________________
8. The magnetic field of a refrigerator pulls the magnet towards the door ________________

What does a vector look like?

There are quite a few ways to represent a vector in word problems. You can see vectors written as **A**, ***A***, $\overline{A}$,or $\vec{A}$. All of these are talking about Vector A. Visually, you draw a vector as an arrow



The tail of the vector is the starting point of the vector or the part of the vector opposite the direction it is going. The head of the vector is the arrowhead that represents the direction the vector is going in. The magnitude of the vector is proportional to the length of the vector. For example, look at the two vectors below. One represents 50mph East and the other represents 25 mph West. Notice how the heads of the two vectors are going in opposite directions and their lengths are proportional to their magnitude of 50 and 25

Draw each of the following vectors with proper direction and scale relative to each's magnitude: 10 meters North, 5 meters West, 20 meters South, 15meters East

Drawing Vectors Practice Problems

Draw each of the following vector pairs with proper direction and scale relative to each other if possible.

1. 10 miles east and 30 miles west

2. 6 meters per second north and 12 meters per second south

3. 4 gallons and 20 gallons

4. 15 N force east and 20 N force east

5. 2 mph east and 10 mph south

6. 9 N force north and 9 N force south

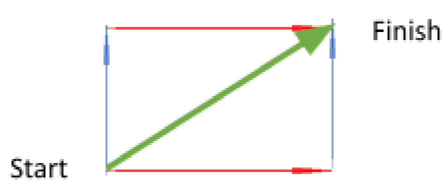
7. 1500 meters west and 1000 meters south

8. 7 mph and 28 mph

How to Add Vectors

You can add two or more vectors together. The Communicative Law of Addition that applies to adding numbers also applies to adding vectors. The Communicative Law of Addition states that changing the order of numbers being added had no impact on the final sum. For example, 5+7=12 and 7+5=12 The same can be said for vectors. The order in which two vectors are added does not matter. If you travel 5 feet forwards and then 7 feet right, you will end up in the same place as you would if you were to walk 7 feet right and then 5 feet forwards.

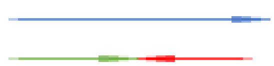
Below are two vectors: Vector A (red/right) and Vector B (blue/up). The communicative law tells us that A + B = B + A.



Vectors A and B are added together and produce Vector C, the green arrow. This vector that connects Vector A and Vector B is the resultant vector. **Resultant vectors** are vectors formed from the sum or two or more vectors.

It is very important to note how the vectors are connected. Vector A and Vector B are connected "head to tail" with the head of Vector A touching the tail of Vector B on the left/top of the diagram above. Similarly, the head of Vector B is touching the tail of Vector A on the right/bottom of the diagram above. All vectors that are added are connected head- -to-tail. It does not matter if you are adding two, two dozen, or two thousand vectors: all vectors added are connected head to tail. Vectors that are connected head-to-head or tail-to-tail are connected incorrectly and can produce very wrong results. When the resultant vector is drawn, its tail (aka starting point) starts at the tail (starting point) of the first vector and its head (aka final point) connects to the head of the final vector being added (finishing point)

Another important thing with vector addition is vectors heading in exact opposite directions. For example, a you can encounter a vector heading East and a vector heading West that are being added together. If the vectors are heading in opposite directions like this, you can subtract the smaller vector from the larger vector to get the resultant.

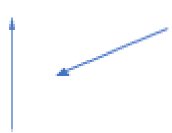


In this case, Vector A (blue/east) and Vector B (red/west) would technically overlap when connected head-to tail but can be drawn like this with Vector C (green) being the resultant

Vector Addition Practice Problems

Add each of the following pairs of vectors. Make the resultant vector easy to see (aka highlight it, label it, draw it in a different color)

1. 

2. 

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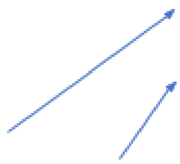

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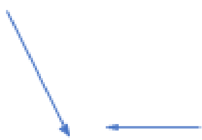

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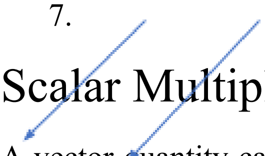


Scalar Multiplication of a Vector

A vector quantity can be multiplied by a scalar (any number) to produce a different vector. If we have Vector A and a scaler "x", we can call our scalar multiple vector xA

The magnitude of xA = $|x|$ (magnitude of A)

The direction of xA remains the same if x is positive and flips if x is negative.

What does this look like? The vectors below show A, $-A$, $2A$, $-2A$, $\frac{1}{2}A$, and $-\frac{1}{2}A$

$2A$

A

$\frac{1}{2}A$

$-A$

$-2A$

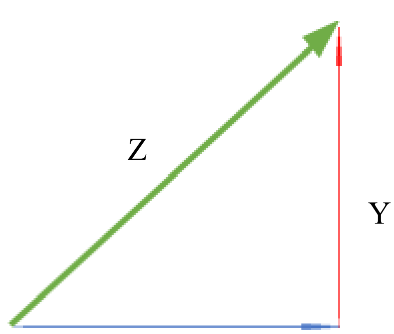
$-\frac{1}{2}A$

Scalar Multiplication Practice Problems

Draw a vector **X**. Then to scale draw vectors -**X**, 3**X**, 5**X**, -3**X**, -5**X**, $\frac{1}{2}$**X**, -$\frac{1}{2}$**X**,

Vector Components

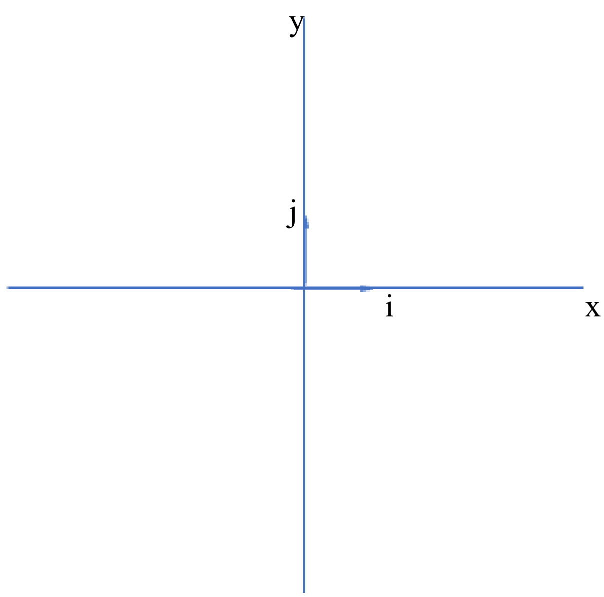
So far, we have talked about how to add vectors by connecting them head – to – tail by drawing them. This is good if we want to get a visual representation of what is going on. However, this is not great for giving the most precise answers. Even the best artists will not be able to draw every vector to scale with proper magnitude and direction. This bring us to the idea of vectors that are two-dimensional. A two-dimensional vector is a vector that lies in a horizontal plane (think x and y axis) and therefore has both a horizontal and vertical component. To put it in terms we already talked about, resultant vector **Z** is equal to the vertical component vector **Y** plus the horizontal component vector **X**.

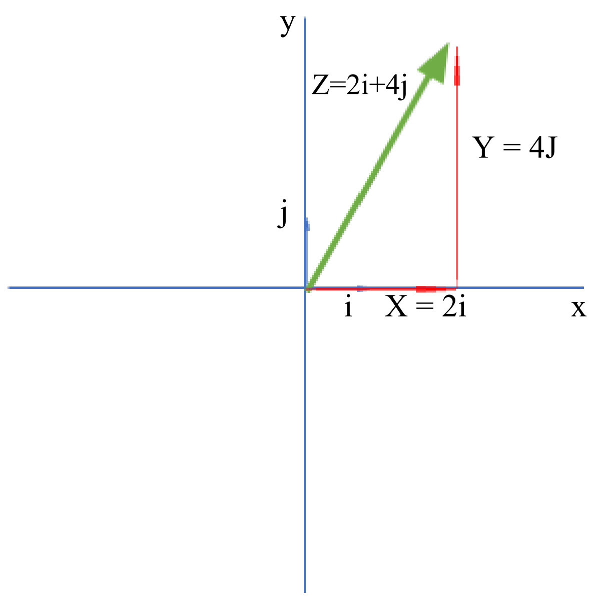


X

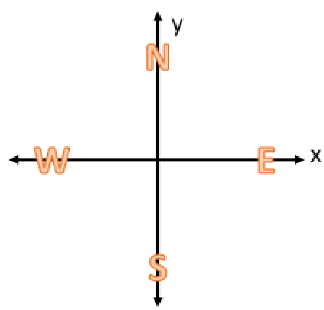
When we encounter a vector that isn't directly on an x-axis or y-axis (a vector directly on an axis will be referred to as "due" N/S/E/W later on), we can break that vector down into its x and y components. This part we will cover in this paragraph is something you may or may not see in college. I asked around and about 1 of my friends had done this in college physics. That being said, College Board puts it on the AP Test so its getting covered. The horizontal aspect of a two-dimensional vector is a scalar multiple of the **horizontal basis vector, i**. The vertical aspect of a two-dimensional vector is a scalar multiple of the **vertical basis vector, j**. These two basis vectors have a magnitude of 1 and are referred to as **unit vectors**. On the AP test, you can see the unit vectors with arrows that appear like "^" above them instead of the usual dots you see above i's and j's.

If we look on an x-and y- axis, we can also label the x-axis the i axis and the y- axis the j axis. Two – dimensional vector, vector **Z**, on the coordinate grid is made up of horizontal vector **X** and vertical vector **Y**.





The name of this section was vector components. In the example above, vectors **X** and **Y** are the **vector components** or **component vectors** of vector **Z**. Earlier on, it was mentioned that j and i are the unit vectors. If these unit vectors have gone from **i** and **j** to 2**i** and 4**j**, this means that they have been multiplied by scalars (similar to the scalar multiplication like X, 3X, 1/2X, etc we did before.) This makes the numbers 2 and 4 in this example the scalar components of vector **Z**. This is why vector **Z** can be written as $Z_x\mathbf{i} + Z_y\mathbf{j}$ where Z_x and Z_y are the scalar components of **Z** with Z_x being the horizontal scalar component and Z_y being the vertical scalar component.



North, south, east, and west can be placed on the coordinate grid. North and East are considered positive (+ i and + j). South and West are considered negative (- i and -j). These signs are very important when dealing with vector components.

Vector Component Practice Problems

Draw each of the following vectors (remember to draw them to scale) and properly label the components as positive or negative i or j. Also draw in the resultant vector and label it properly using i and j

1. 5 m north and 10 m west

2. 4 miles east and 6 miles north

3. 20 miles south and 5 miles east

4. 12 mph west and 36 mph north

5. 40 N south and 20 N west

6. 30 feet east and 100 feet north

Adding, Subtracting, and Multiplying with Vector Components

Lets say we have two vectors, vector X and vector Y. We can use their components to add, subtract, and multiply.

To add two vectors using their components, you add their x and y components respectively.

$$\mathbf{X} + \mathbf{Y} = (X_x + Y_x)i + (X_y + Y_y)j$$

Say we have vectors **C** = 4i + 6j and **G**= -1i + 5j. To solve **C**+**G** we would do the following steps:

$$\mathbf{C} + \mathbf{G} = (C_x + G_x)i + (C_y + G_y)j$$

$$\mathbf{C} + \mathbf{G} = (4 - 1)i + (6 + 5)j$$

$$\mathbf{C} + \mathbf{G} = 3i + 11\ j$$

To subtract two vectors using their components, you subtract their x and y components respectively.

$$\mathbf{X} + \mathbf{Y} = (X_x - Y_x)i + (X_y - Y_y)j$$

Say we have vectors **T** = 5i + 6j and **D**= 2i + 1j. To solve **T - D** we would do the following steps:

$$\mathbf{T - D} = (T_x - D_x)i + (T_y + D_y)j$$

$$\mathbf{T - D} = (5 - 2)i + (6 - 1)j$$

$$\mathbf{T - D} = 3i + 5j$$

To do scalar multiplication of a two-dimensional vector using its components, you multiply each component by "k" the scalar.

$$k\mathbf{A} = (kA_x)i + (kA_y)j$$

Say we have vector $\mathbf{Z} = 7i + 4j$ and we multiply it by scalar $k = 3$.

$$k\mathbf{Z} = (kZ_x)i + (kZ_y)j$$

$$3\mathbf{Z} = (3x7)i + (3x4)j$$

$$3\mathbf{Z} = 21i + 12j$$

It is important to note that if you have a problem where you have vectors A and B and you are asked to solve for 5A – 2B, P.E.M.D.A.S. applies and you must carry out the multiplication of each individual vector and its components before completing the addition or subtraction step

Adding, Subtracting, and Multiplying with Vector Components Practice Problems.

For each of the following problems, complete the math problem using the following vectors:
A: 2i + 7j, **B**: -4i + 9j, **C**: 3i + 6j, **D**: -8i + -1j

1. **A** + **B**

2. **C**+ **A**

3. **B** – **D**

4. **D** – **A**

5. 4**D**

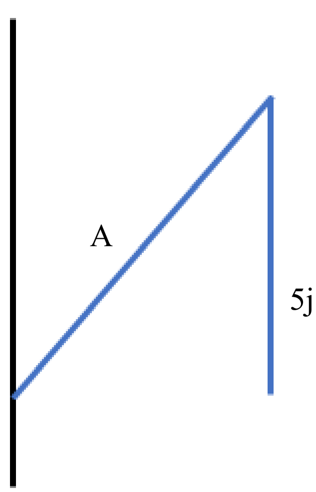
6. -2**C**

7. 7**B** – 2**A**

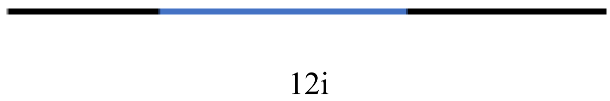
8. 3**A** + 5**C**

Using Vector Components to find the Magnitude of a Vector

There will be many times in physics that you will have to find the magnitude of a resultant vector by using its two component vectors. This can be done by using the Pythagorean Theorem $a^2 + b^2 = c^2$. If you have a vector that is written in the format of $A = A_x i + A_y j$, the equation for solving for the magnitude will be $A = \sqrt{(A_x)^2 + (A_y)^2}$. This is just a rewritten form of the Pythagorean Theorem using the magnitude of the vector A as the hypotenuse and vector components A_x and A_y as the two sides of the triangle.

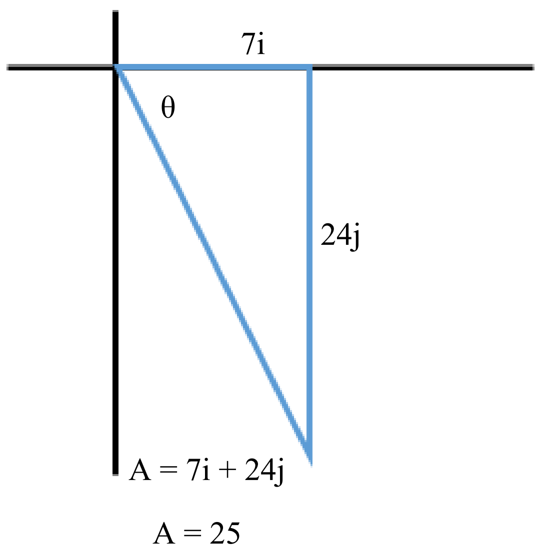


In this example, the magnitude of A can be solved using $A = \sqrt{5^2 + 12^2} = 13$

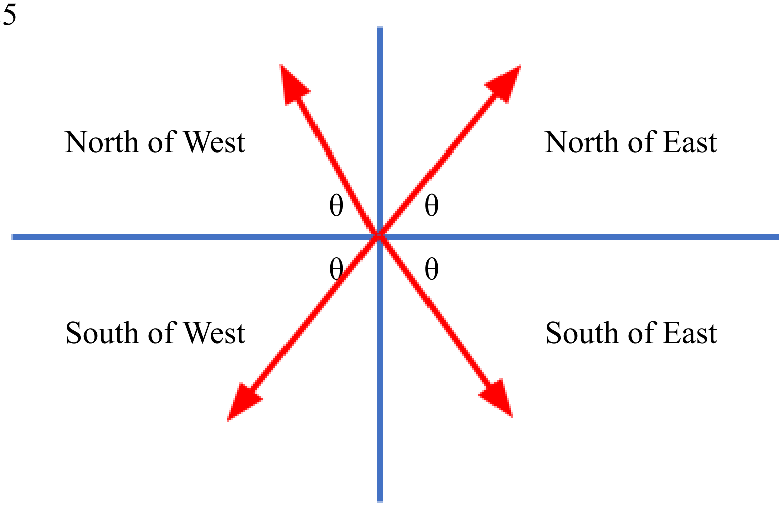


Solving for The Direction of a Vector

Once you have calculated the magnitude of a vector from its components, you will have to find its direction. This is because a vector has both magnitude and direction – if you only find one, you've only found half your answer. To calculate the angle of a vector, you will use the inverse tan function. This means the formula to solve for the angle of a vector is $tan^{-1}\left(\frac{opp}{adj}\right) = \ \theta$



In this example, the magnitude of A can be found using the Pythagorean Theorem. To find the direction, we use $tan^{-1}\left(\frac{opp}{adj}\right) = \ \theta$ and get 73.7°



For naming directions, we will use the angle the vector makes with the x-axis unless stated otherwise. THIS MEANS WE WILL NEVER (YES NEVER) USE WEST OF NORTH, EAST OF NORTH, WEST OF SOUTH, OR WEST OF EAST IN THIS ROOM. You have been warned.

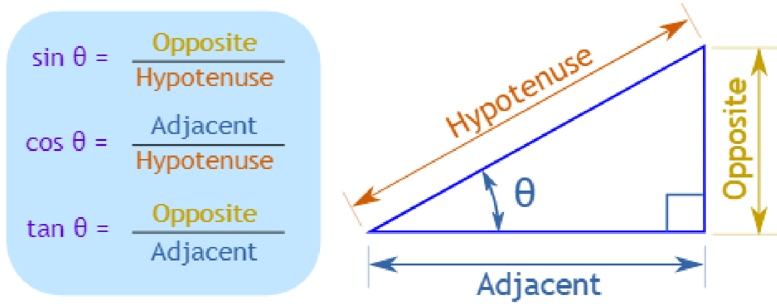
X- Component	Y- Component	Direction
+	+	North of East

-	+	North of West
-	-	South of West
+	-	South of East

It is important to note the positive and negative signs apply only to directions and do not need to be put into the Pythagorean Theorem or inverse tan function.

Using Sin, Cos, and Tan to Calculate Vector Components

For many of the problems we will do (such as problems involving projectiles or slopes), we will have to break a vector down into its x and y components. To do this, we can use the trig functions of sin, cos, and tan (although tan is not really necessary for solving for a side, it never hurts to review.)



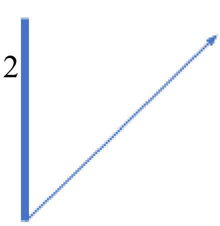
For physics purposes, we will always look at the Hypotenuse as the magnitude of the vector. The angle θ will always be the angle the vector makes with the x-axis unless stated otherwise. This means that **the X component will always be adjacent and can be solved by using Hyp x cos(θ)** and the **Y component will always be opposite and can be solved by using Hyp x sin (θ)**.

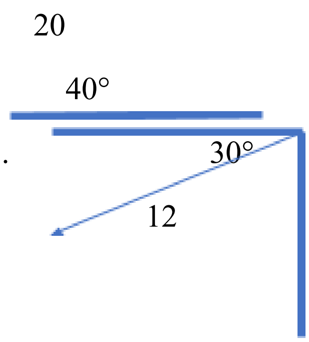
Vector Magnitude & Sin, Cos, Tan Practice Problems

If you are given a vector and its angle, use trig functions to calculate the x and y components. If you are given the x and y components, use the Pythagorean Theorem and trig functions to find the vector's magnitude and direction

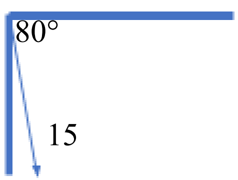
1. 4i and 5j

2



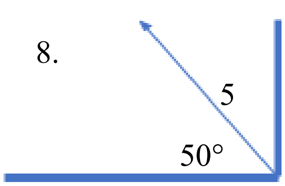
3. 

4. -6i and 3j

5. 

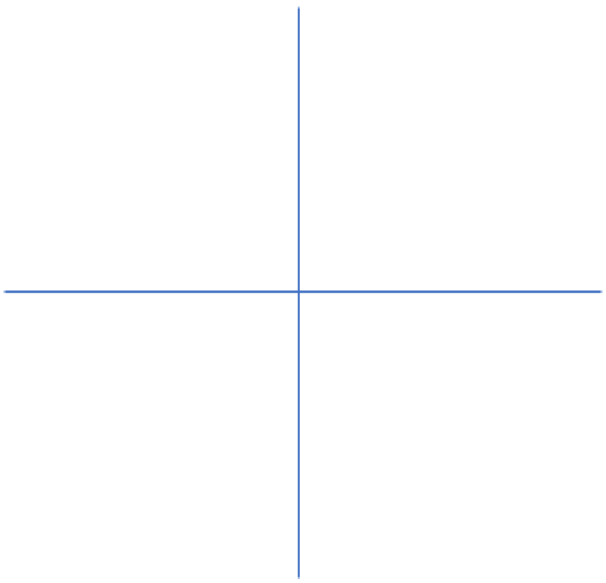
6. -1i and -7j

7. 9i and -2j

8. 

9. 3i and 8j

10. Draw each of the following vectors. Then, break them into their x and y components. Finally, add them all together and find the resultant vector's magnitude and direction. Vectors: 10m due North, 6m 40° N of W, 15m 20° S of E, and 5m 80° S of W.



11. Add the following vectors and find both the magnitude and direction of their resultant. (FYI, if you are every working with vectors, you need a magnitude and direction even if you aren't asked for it. 15 miles north, 10 miles 30 degrees south of west, 4 miles 75 degrees north of east.

12. Add the following vectors. 100 feet east, 20 feet south, 45 feet west, 20 feet north, 35 feet 45 degrees north of west, 75 feet 60 degrees south of east.